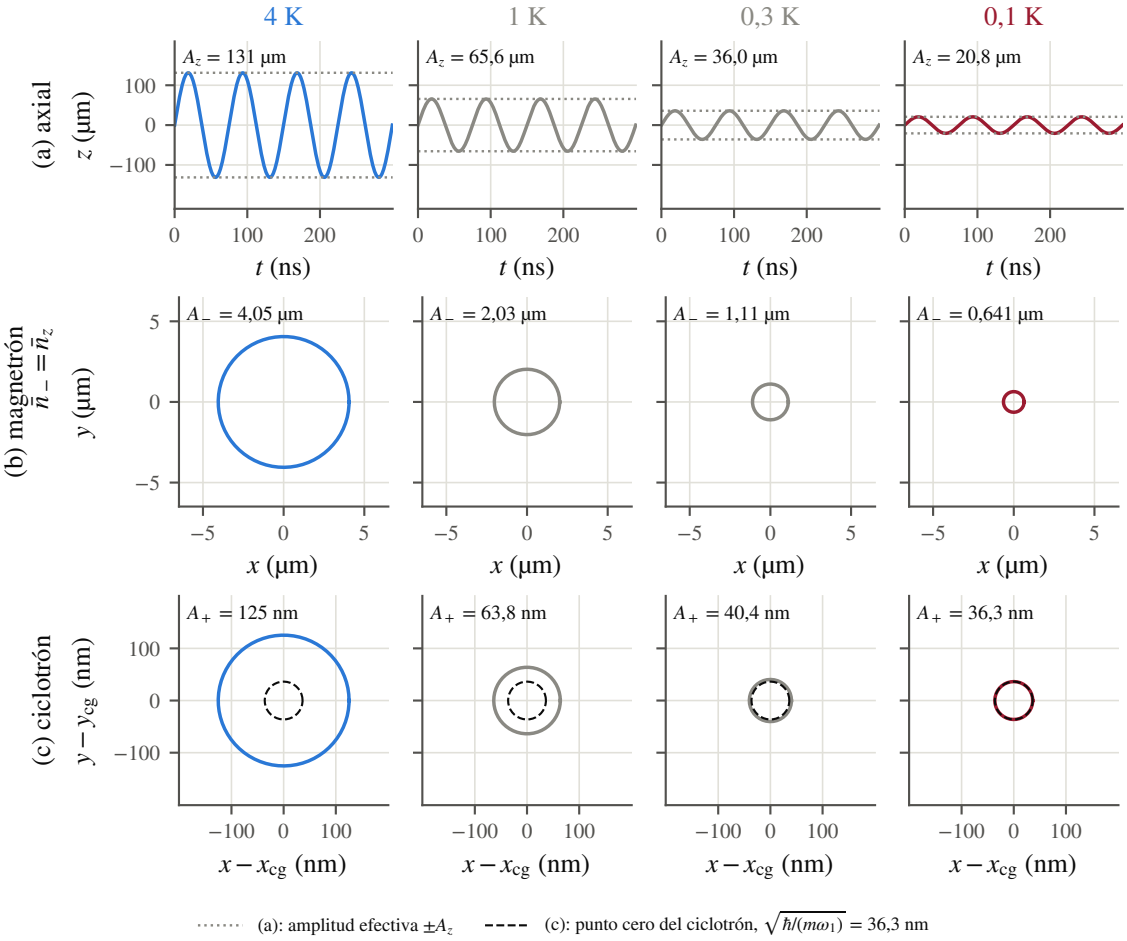
4 K
1 K
0,3 K
0,1 K
(a) axial
z (μm)
A_z = 131 μm
A_z = 65,6 μm
A_z = 36,0 μm
A_z = 20,8 μm
100
0
−100
0
100
200
t (ns)
(b) magnetrón
$\bar{n}_- = \bar{n}_z$
y (μm)
A_- = 4,05 μm
A_- = 2,03 μm
A_- = 1,11 μm
A_- = 0,641 μm
5
0
−5
x (μm)
(c) ciclotrón
$y - y_{cg}$ (nm)
A_+ = 125 nm
A_+ = 63,8 nm
A_+ = 40,4 nm
A_+ = 36,3 nm
$x - x_{cg}$ (nm)
(a): amplitud efectiva $\pm A_z$
(c): punto cero del ciclotrón, $\sqrt{\hbar/(m\omega_1)}$ = 36,3 nm